PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE D · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

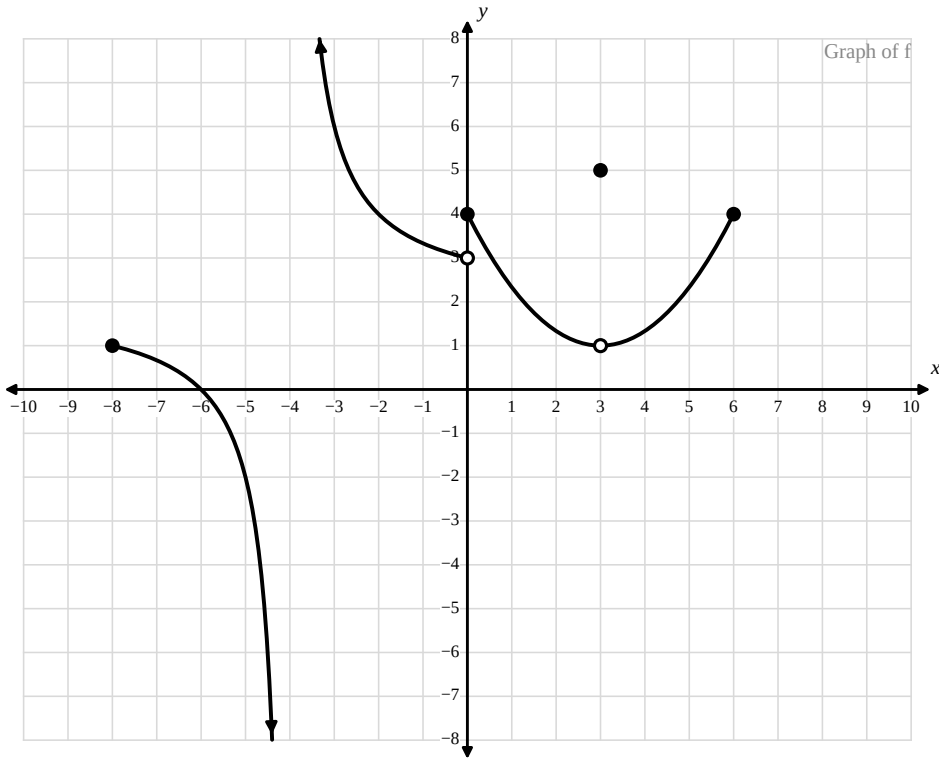
_____

_____

Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

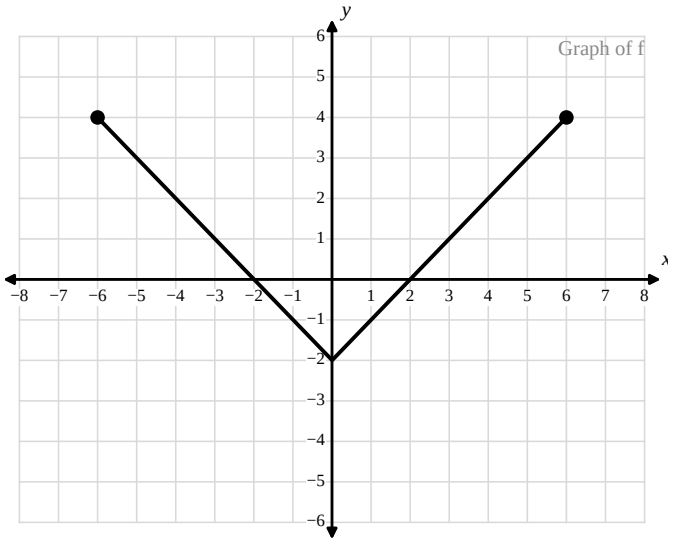


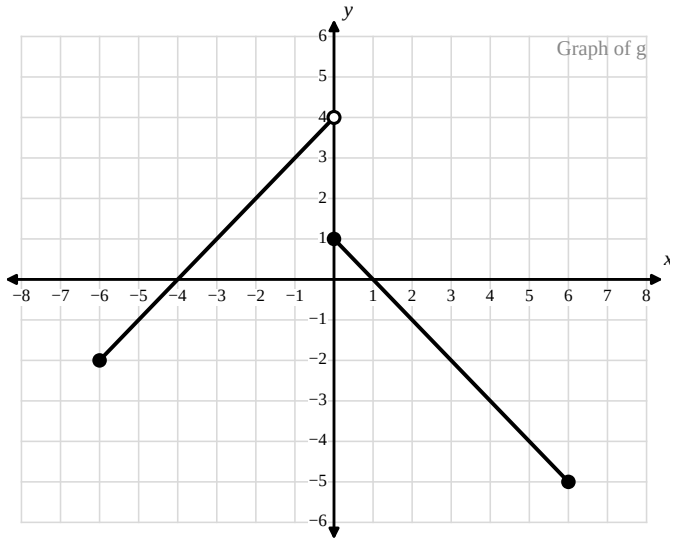
1. $\lim_{x\to 3} f(x) =$	**2.** $\lim_{x\to -4^-} f(x) =$	**3.** $\lim_{x\to 0} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 4$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim\limits_{x \to 0^-} \dfrac{f(x)}{g(x)}$

(2 points)

6. Use the table to estimate $\lim\limits_{x \to 0} k(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−0.1	−0.01	−0.001	→ 0 ←	0.001	0.01	0.1
$k(x)$	0.951626	0.995017	0.999500		1.000500	1.005017	1.051709

$\lim\limits_{x \to 0^-} k(x) \approx$ _____ $\lim\limits_{x \to 0^+} k(x) \approx$ _____ $\lim\limits_{x \to 0} k(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim\limits_{x\to\frac{\pi}{3}} (6\cos x - \tan x)$	8. $\lim\limits_{x\to -2} \dfrac{x^3 - 4x}{x^2 + 2x}$
9. $\lim\limits_{h\to 0} \dfrac{\sqrt{25+h} - 5}{h}$	10. $\lim\limits_{x\to 1} \dfrac{\frac{2}{x+1} - 1}{x - 1}$
11. For $f(x) = 3x^2 + x$, find $\lim\limits_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$	

PCBC H Name **ANSWER KEY** **22** /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

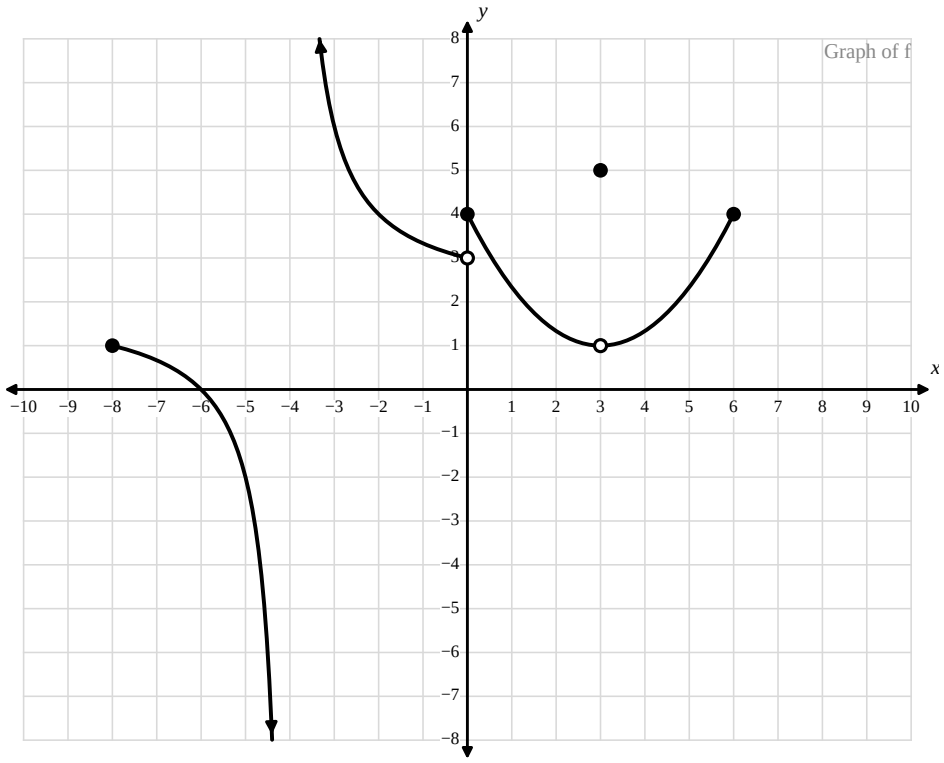
PRACTICE D · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)



1. $\lim_{x\to 3} f(x) =$

Both sides head to the open circle at $(3, 1)$.

1

2. $\lim_{x\to -4^-} f(x) =$

Just left of $x = -4$ *the curve falls without bound.*

$-\infty$

3. $\lim_{x\to 0} f(x) =$

$\lim_{x\to 0^-} f(x) = 3$, $\lim_{x\to 0^+} f(x) = 4$

$3 \neq 4$, *so* $\lim_{x\to 0} f(x)$ *does not exist.*

DNE

(2 points)

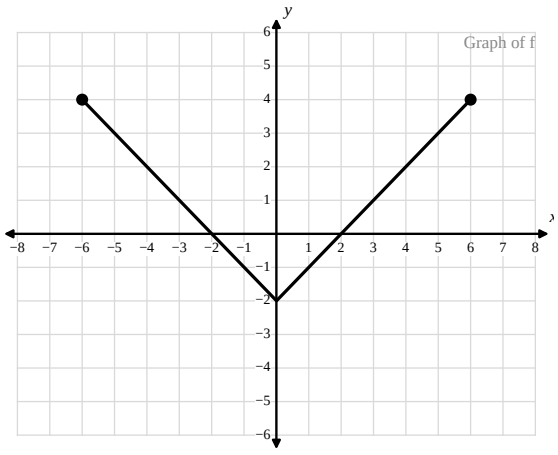
4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 4$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

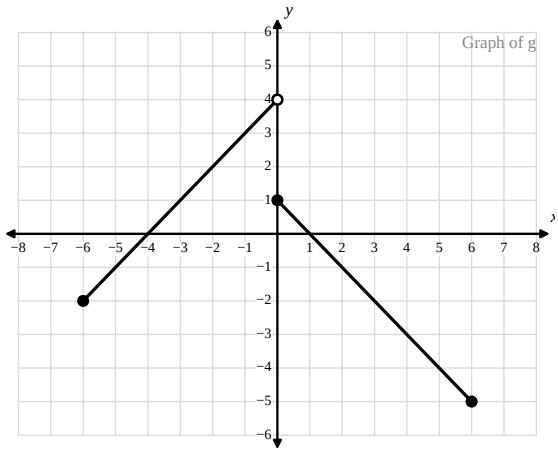
Height 4 is reached from both sides only at $x = -2$.

Not $x = 0$ *(right side only; the left heads to 3) and not* $x = 6$ *(the graph ends there).*

$a = -2$

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to 0^-} \dfrac{f(x)}{g(x)}$

$$\lim_{x \to 0^-} \frac{f(x)}{g(x)} = \frac{\lim_{x \to 0^-} f(x)}{\lim_{x \to 0^-} g(x)}$$

$$= \frac{-2}{4} = -\frac{1}{2}$$

$-\frac{1}{2}$

(2 points)

6. Use the table to estimate $\lim_{x \to 0} k(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−0.1	−0.01	−0.001	→ 0 ←	0.001	0.01	0.1
$k(x)$	0.951626	0.995017	0.999500		1.000500	1.005017	1.051709

Left: 0.951626, 0.995017, 0.999500

Right: 1.051709, 1.005017, 1.000500

Both sides agree. A table estimates; it does not prove.

≈ 1

$\lim_{x \to 0^-} k(x) \approx$ 1 $\lim_{x \to 0^+} k(x) \approx$ 1 $\lim_{x \to 0} k(x) \approx$ 1

Evaluate each limit. (2 points each)

7. $\lim\limits_{x\to\frac{\pi}{3}} (6\cos x - \tan x)$

$$\lim_{x\to\frac{\pi}{3}} (6\cos x - \tan x) = 6\cos\frac{\pi}{3} - \tan\frac{\pi}{3}$$
$$= 6\left(\frac{1}{2}\right) - \sqrt{3} = 3 - \sqrt{3}$$

$3-\sqrt{3}$

8. $\lim\limits_{x\to -2} \dfrac{x^3 - 4x}{x^2 + 2x}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to -2} \frac{x^3 - 4x}{x^2 + 2x} = \lim_{x\to -2} \frac{x(x-2)(x+2)}{x(x+2)}$$
$$= \lim_{x\to -2} (x-2) = -2 - 2 = -4$$

-4

9. $\lim\limits_{h\to 0} \dfrac{\sqrt{25+h} - 5}{h}$

Substituting $h = 0$ produces 0/0.

$$\lim_{h\to 0} \frac{\sqrt{25+h} - 5}{h} = \lim_{h\to 0} \frac{(25+h) - 25}{h\left(\sqrt{25+h} + 5\right)}$$
$$= \lim_{h\to 0} \frac{1}{\sqrt{25+h} + 5} = \frac{1}{5+5} = \frac{1}{10}$$

$\frac{1}{10}$

10. $\lim\limits_{x\to 1} \dfrac{\frac{2}{x+1} - 1}{x - 1}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to 1} \frac{\frac{2}{x+1} - 1}{x - 1} = \lim_{x\to 1} \frac{2 - (x+1)}{(x+1)(x-1)}$$
$$= \lim_{x\to 1} \frac{-(x-1)}{(x+1)(x-1)} = \lim_{x\to 1} \frac{-1}{x+1}$$
$$= \frac{-1}{1+1} = -\frac{1}{2}$$

$-\frac{1}{2}$

11. For $f(x) = 3x^2 + x$, find $\lim\limits_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$

$$\lim_{h\to 0} \frac{\left[3(x+h)^2 + (x+h)\right] - \left(3x^2 + x\right)}{h}$$
$$= \lim_{h\to 0} \frac{3x^2 + 6xh + 3h^2 + x + h - 3x^2 - x}{h}$$
$$= \lim_{h\to 0} \frac{h(6x + 3h + 1)}{h} = \lim_{h\to 0} (6x + 3h + 1)$$
$$= 6x + 0 + 1 = 6x + 1$$

$6x+1$